NOT MEASUREMENT SENSITIVE

MIL-T-62102F
17 October 1990
SUPERSEDING
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9 March 1987

MILITARY SPECIFICATION

* TRUCK TRACTORS: SEMITRAILER SPOTTER, WITH HALF CAB, DIESEL AND GASOLINE ENGINE DRIVEN, 32,000 GVW, 55,000 GCW AND 44,000 GVW, 90,000 GCW, 4x2, MODIFIED COMMERCIAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers half cab, diesel and gasoline engine driven, four-wheel, two-rear-wheel drive modified commercial truck tractors used for spotting semitrailers. The truck tractors have a minimum gross vehicle weight (GVW) of 32,000 or 44,000 pounds under normal operating conditions. Vehicles procured under this specification are commercial items which are warranted by the manufacturer as specified in acquisition documents.

* 1.2 Class. The class of the vehicle shall be determined by its minimum gross vehicle weight rating as follows:

Class 32 - 32,000 pounds GVW
Class 44 - 44,000 pounds GVW.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-UED, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC-2320

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* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

	W-B-131	- Battery, Storage: Vehicular, Ignition, Lighting and Starting.
*	VV-F-800	- Fuel Oil, Diesel.

MILITARY

	MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
*	MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
*	MIL-T-83133	- Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.

STANDARDS

FEDERAL

*	FED-STD-297	- Rustproofing of Commercial (Nontactical) Vehicles.

MILITARY

*	MIL-STD-1223	- Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking, and Data Plate Standards.
	MS51118	- Pintle Assembly, Towing - 40,000 Lbs. Capacity, Manual Release.
	MS75020	- Connector, Plug, Electrical - 12-Contact, Intervehicular, 28-Volt, Waterproof.
	MS75021	- Connector, Receptacle, Electrical - 12 Contact, Intervehicular, 28-Volt, Waterproof.

* (Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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* DEPARTMENT OF DEFENSE
Department of Defense Index of Specifications and Standards (DODISS).

(Copies of the DODISS are available on a yearly subscription basis either from the Government Printing Office for hard copy, or microfiche copies are available from the Director, Navy Publication and Printing Service Office, 700 Robbins Avenue, Philadelphia, PA 19111-5093.)

DEPARTMENT OF TRANSPORTATION (DOT)
Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.

(Application for copies of DOT publications should reference the code of Federal Regulations, 49 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)
Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.
Noise Emission Standards for Transportation Equipment - Medium and Heavy Trucks.

(Application for copies of EPA publications should reference the Code of Federal Regulations, 40 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the Government documents cited in the solicitation. (See 6.2).

* THE EUROPEAN TYRE AND RIM TECHNICAL ORGANISATION (ETRTO)
Standards Manual

(Application for copies of the ETRTO publications should be addressed to the European Tyre and Rim Technical Organisation, 32, Avenue Brugmann, 1060 Brussels, Belgium.)

SAE, INC.
SAE Standards and Recommended Practices
J318 - Air Brake Gladhand Service (Control) and Emergency (Supply) Line Couplers - Trucks, Truck-Tractors, and Trailers (DOD adopted).

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J350	- Spark Arrester Test Procedure for Medium Size Engines (DOD adopted).
J537	- Storage Batteries.
J551	- Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (30-1000 MHz).
J560	- Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
J588	- Turn Signal Lamps.
J589	- Turn Signal Switch.
J682	- Rear Wheel Splash and Stone Throw Protection (DOD adopted).
J688	- Truck Ability Prediction Procedure (DOD adopted).
J700	- Fifth Wheel Kingpin - Commercial Trailers and Semitrailers (DOD adopted).
J844	- Nonmetallic Air Brake System Tubing (DOD adopted).
J1067	- Seven-Conductor Jacketed Cable for Truck-Trailer Connections.
J1349	- Engine Power Test Code - Spark Ignition and Diesel.

(Application for copies of SAE publications should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.
Year Book.

* (Application for copies of Tire and Rim Association publications should be addressed to The Tire and Rim Association, Inc., 175 Montrose West Ave., Copley, OH 44321.)

* (Non-Government standards and other publications are normally available from the organizations that prepare or distribute the document. These documents also may be available in or through libraries or other informational services).

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.9, the vehicle, components, assemblies, and accessories to be delivered under the contract shall be standard or optional items, which meet or exceed the requirements of this specification. Except as specified in

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3.1.1 through 3.1.1.9, no removal, substitution or alteration of the chassis manufacturer's standard or optional chassis model components shall be made. All chassis items shall be as represented in the chassis manufacturer's technical data book. Special bodies or mounted equipment shall be as represented in the body and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material, identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be available to the engineering offices of the procuring activity, prior to delivery of the items. The chassis model furnished shall be not older than the chassis manufacturer's current model on the date of invitation for bids.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

* 3.1.1.1 Treatment and painting. The vehicle shall be treated and painted in accordance with MIL-STD-1223. As specified, by the procuring activity for the appropriate military service (see 6.2), the exterior color shall be in accordance with MIL-STD-1223.

* 3.1.1.2 Markings and data plates. As specified by the procuring activity for the appropriate military service (see 6.2), identification marking and data plates shall be in accordance with MIL-STD-1223. When specified (see 6.2), concealed markings shall be furnished.

3.1.1.3 Rustproofing. When specified (see 6.2), the vehicle shall be rustproofed in accordance with FED-STD-297. When specified (see 6.2), tropical rustproofing in accordance with FED-STD-297 shall be furnished.

3.1.1.4 Drain plug. The drain plug for the rear axle differential shall be of the permanent magnet type.

3.1.1.5 Towing devices. Not less than two hooks, loops, pins, or a center mounted eye for towing the vehicle shall be furnished on the front of the vehicle and not less than two hooks, loops, or pins for towing the vehicle shall be furnished on the rear of the vehicle. The towing devices shall be frame mounted. Front towing devices shall have a total capacity of not less than 60,000 pounds and rear towing devices shall have a total capacity of not less than 35,000 pounds.

3.1.1.6 Trailer towing package. A trailer towing package consisting of a pintle, safety chain attachment devices, a lighting receptacle and associated reinforcements and wiring shall be installed on the rear of the vehicle. The pintle shall be of the rotating type conforming to MS51118. The pintle shall be installed on the chassis frame with reinforcements to transfer a vertical tongue load of not less than 4,000 pounds and a horizontal drawbar load of not less than 40,000 pounds directly to the

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chassis rails. The rearmost portion of the pintle shall be not more than 4 inches forward of the rearmost part of the vehicle. Two trailer safety chain attachment devices, one adjacent to each side of the pintle, shall be provided. Each attachment device shall provide an ultimate strength of not less than 40,000 pounds and shall be capable of accommodating a standard grab hook (4-9/16 inches wide, 1-3/16 inches thick, 25/32-inch throat width) for a 5/8-inch chain. A lighting receptacle, conforming to SAE J560 with its conductors connected and color coded as specified therein, shall be mounted in a readily accessible location near the pintle.

3.1.1.7 Front fenders. Fenders shall be furnished over that portion of the front wheel assemblies not covered by the cab. The front, top, and the rear of the wheel assemblies shall be enclosed. Sufficient reinforcement shall be furnished to minimize vibration. Removable rubber splash shields shall be furnished at the rear of the fenders.

3.1.1.8 Wheel splash and stone throw protection. The vehicle shall have rigid half fenders over the rear wheels and removable rubber mud flaps to the rear of the rear wheels. The mud flaps and their extension supports shall be readily removable, to increase landing wheel clearance, without the use of hand tools. The half fenders need extend down only to the height of the centerline of the rear axle. A metal strip, not less than 0.125 inch thick and not less than one inch wide, extending the entire width of the mud flap, shall be installed to prevent bolt heads or bolt nuts from damaging the mud flap. As an alternate method of attaching the mud flaps, tabs or clips with minimum surface contact dimensions of 1 inch high by 1.125 inch wide by 0.094 inch thick shall be furnished at each bolt. All tilt cabs shall have rubber mud flaps to the rear of the front wheels. All splash shield and mud flap installations, front and rear, shall conform to the rear wheel splash and stone throw protection provisions of SAE J682.

* 3.1.1.9 Brake lights. At least one pair of brake lights shall override the four-way emergency flasher or the two systems shall be independent of each other. Modifications to the maufacturer's standard product to accommodate this requirement shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard. If additional lights are added to the vehicle, the lights shall be selected from the chassis manufacturer's standard matching hardware. The brake lights need to override the four-way flasher only when coupled to a semitrailer.

3.2 General design.

3.2.1 Federal Motor Vehicle Safety Standards. The vehicle and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

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3.2.2 Air pollution control. The vehicle shall comply with EPA Regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture. In addition, vehicles destined for California shall comply with State of California regulations governing air pollution control in effect on the date of manufacture.

* 3.2.3 Sound level. The interior sound level shall not exceed 84 db(A) when measured in accordance with Federal Motor Carrier Safety Regulation 393.94. The vehicle exterior sound level shall conform to EPA Noise Emission Standards for Transportation Equipment, Medium and Heavy Trucks.

3.2.4 Curb weight. The curb weight shall include the weight of the chassis and cab, with all attachments, accessories, and equipment; fifth wheel; and a full complement of fuel, lubricants and coolant.

3.2.5 Gross vehicle weight. Gross vehicle weight (GVW) shall consist of curb weight, operator weight (computed at 175 pounds), and a payload to provide not less than the specified GVW.

3.2.6 Gross combination weight. Gross combination weight (GCW) shall consist of truck tractor curb weight, operator weight (computed at 175 pounds), and the weight of a semitrailer, loaded to provide not less than the specified GCW. The fifth wheel shall be located so that with the truck tractor loaded to GVW, the load rating of the chassis components are not exceeded.

* 3.2.7 Ratings. Vehicle ratings shall be the manufacturer's published rating. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings must be submitted to the engineering office of the procuring activity. Minimum ratings for class 32 truck tractor shall be 32,000 GVW and 55,000 GCW and for class 44, 44,000 GVW and 90,000 GCW.

3.2.8 Dimensions. The vehicle wheelbase shall be not more than 104 inches and shall provide a cab-to-axle dimension of not more than 84 inches. The overall width of the vehicle, exclusive of mirrors, lights, reflectors and tires, shall be not more than 96 inches. The width measured over the tires shall be not more than 100 inches.

3.2.9 Accessibility. The design of the vehicle and optional equipment shall permit access for routine servicing and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

3.2.10 Asbestos. Except as permitted in 3.4.11, asbestos materials shall not be used in any form in any part of the vehicle.

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3.3 Performance.

* 3.3.1 Speeds and gradeability. High and low speed requirements shall be met with the truck tractor loaded to specified GVW and GCW.

* 3.3.1.1 High speed gradeability. The vehicle shall ascend a continuous grade of 1.0 percent at 35 miles per hour (mph) for size 32 and 12 mph for size 44. Gradeability requirements shall be met with the main transmission in direct drive. Gradeability shall be verified with calculations in accordance with SAE J688 (see 6.3).

3.3.1.2 Low speed. Low speed shall be calculated with the engine operating at not less than 35 percent of the recommended governed speed, using torque converter ratio and the highest numerical ratio in the transmission and shall provide a vehicle speed of not more than 4 mph.

* 3.3.1.3 Maximum geared speed. The maximum geared speed at engine governed speed shall be not less than 45 mph for size 32 and shall be between 12 and 15 mph for size 44. Conformance to geared speed specified shall be determined by calculating in accordance with the following formula:

$$\text{Maximum geared speed (mph)} = \frac{\text{Governed speed (rpm)}}{\text{Total gear reduction x tire factor (see 6.3).}}$$

3.3.2 Service brakes. The service brakes shall control and hold the vehicle, when loaded to its specified gross weight, on a 30 percent grade. The service brakes shall stop the tractor-semitrailer combination, loaded to specified GCW, within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52.

3.4 Chassis components.

3.4.1 Engine. The engine furnished shall be the chassis manufacturer's standard or optional engine for the commercial model truck which meets or exceeds the requirements of this specification.

* 3.4.1.1 Diesel engine. Unless otherwise specified (see 3.4.1.2), the vehicle shall be equipped with a liquid cooled, compression ignition, two-stroke or four-stroke cycle diesel engine, with not less than six cylinders. Engine net horsepower used in performance prediction calculations shall be determined in accordance with SAE J1349. The engine shall demonstrate the performance characteristic specified herein when using the diesel fuel conforming to VV-F-800. In addition, when specified (see 6.2), the engine shall operate satisfactorily on grade JP-4 and grade JP-5 fuel conforming to MIL-T-5624 under emergency, short duration conditions and on grade JP-8 fuel conforming to MIL-T-83133 under normal conditions. A power loss when operating on JP-4, JP-5 or JP-8 is acceptable.

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* 3.4.1.2 Gasoline engine. When specified (see 6.2), the vehicle shall be equipped with a liquid cooled, internal combustion, four-stroke cycle gasoline engine with not less than six cylinders. The engine furnished shall produce the required vehicle performance when operated on unleaded fuel with a research octane rating of 91, at an engine speed not more than the manufacturer's recommended operating speed. The engine shall be capable of warranted operation on unleaded fuel, when used in accordance with the operator's manual. Unless otherwise specified or required by foreign laws, vehicles for overseas use shall be capable of accepting and operating on leaded gasoline. Net horsepower figures used in performance prediction calculations shall be determined in accordance with SAE J1349.

* 3.4.1.3 Oil filter. A full flow or combination full flow and bypass oil filter with replacement element shall be furnished.

3.4.1.4 Governor. An engine governor shall be furnished and set and sealed to limit the engine to engine manufacturer's maximum recommended operating speed.

* 3.4.1.5 Cooling system. The cooling system shall maintain the engine coolant at a temperature below the boiling point with the vehicle loaded to GVW and operated at an altitude of 10,000 feet above sea level or in an ambient air temperature of not less than 125 degrees Fahrenheit (°F). The coolant system shall include a surge tank or a coolant recovery reservoir of not less than two-quart capacity. On tilt cab models, a radiator servicing access door shall be provided to allow verification of the coolant level.

* 3.4.1.6 Coolant temperature control. Thermostatic control of engine coolant temperature shall be provided. On diesel engine driven vehicles, the control shall include complete thermostatic control of all coolant flow through the radiator.

* 3.4.1.7 Fan clutch. A fan clutch shall be provided. The fan clutch shall reduce the fan speed automatically when the fan is not required for engine cooling. The fan clutch shall be asbestos free.

3.4.1.8 Silicone rubber hoses. When specified (see 6.2), silicone rubber radiator and heater hoses shall be furnished.

* 3.4.1.9 Power plant heaters and fuel warmer. When specified (see 6.2), a coolant heater, an engine oil heater, and a fuel warmer (diesel engine driven vehicles only) shall be provided. Unless otherwise specified (see 6.2), a battery heater shall be provided when power plant heaters are specified. Heaters shall operate on 110-volt alternating current (ac), and shall be wired through a junction block, including fuse or circuit breaker, to a single three-pronged (male), weatherproof, slave receptacle for receiving external power and for grounding the vehicle. A three-wire connecting cable, 25 feet long and of adequate line capacity to supply power

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for all heater units simultaneously, shall be furnished. The connecting cable shall include a matching female connector at the vehicle end and a standard three-pronged (two power plus one ground) male connector at the other end. Electrical apparatus shall conform to Federal Motor Carrier Safety Regulation 393.77(c)(7). Electrical insulation of connecting cable shall withstand normal operating stresses in low ambient air temperature (down to minus 60°F) without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with vehicle component operation and without loose excess wire. Provisions for stowage of the cable shall be provided in the vehicle cab. Heaters shall be furnished as follows:

(a) Coolant heater, 1500-watt minimum rating, shall be installed in the engine block, or lower coolant inlet hose. Engine thermostat with an operating range of 170° to 195°F shall be installed.

* (b) An oil pan heater of the permanent external surface mount, permanent in-pan mount, or immersion type that meets the following requirements, shall be installed:
  (1) Not less than 22 watts per quart heating capacity
  (2) Not more than 18 watts per square inch heating capacity
  (3) Thermal balance design or thermostat control providing for uninterrupted operation
  (4) Provision for mounting below minimum service oil level.

(c) Battery heater shall have a capacity adequate to maintain the battery electrolyte at a temperature of not less than 10°F during vehicle exposure in ambient air temperature as low as minus 60°F, and shall embody a thermostat to limit the temperature of the electrolyte to not more than 80°F.

(d) For diesel engine driven vehicles, a fuel warmer or preheater shall be provided to prevent clogging of fuel filters due to wax crystallization in the fuel. The fuel warmer shall use engine coolant to transfer sufficient heat to the diesel fuel to heat it from an inlet temperature of minus 40°F to an outlet temperature of plus 9°F, with a flow rate not less than the maximum fuel demand of the engine fuel system. A coolant shutoff valve shall be provided for the coolant inlet side of the fuel warmer unit.

3.4.2 Electrical system. The electrical system shall be in accordance with Federal Motor Carrier Safety Regulations 393.27 through 393.31 and 393.33.

* 3.4.2.1 Starting system. For diesel engine driven vehicles, a 12- or 24-volt direct current (dc) starting system, with a 12-volt dc lighting system shall be furnished. Engine starting system shall include a glow plug, an ether system or electric grid heater. If an ether system is furnished in lieu of a glow plug or electric grid heater, it shall be of the measured shot type. The measured shot type ether system shall be key operated or manually operated from the driver's compartment, and shall be inoperative with the

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engine warm. Complete provisions for a replaceable ether reservoir of not less than 12 fluid ounces shall be furnished. A reservoir need not be furnished.

* 3.4.2.2 Ignition system. For gasoline engine driven vehicles, a 12-volt dc ignition system shall be furnished.

* 3.4.2.3 Alternator. Unless otherwise specified (see 6.2), on diesel engine driven vehicle, a minimum 75-ampere alternator shall be furnished. The alternator output with the engine at engine idle speed shall be not less than 45 amperes. Unless otherwise specified (see 6.2) on gasoline engine driven vehicles, an alternator of not less than 85 amperes rated capacity, and which provides not less than 45 amperes dc output at normal engine idle speed, shall be furnished.

3.4.2.4 Lighting. All vehicle lights, reflectors, and wiring shall be as specified herein and shall conform to Federal Motor Carrier Safety Regulations 393.12, 393.13, 393.19, 393.20 and 393.22 through 393.26(d).

3.4.2.5 Turn signals. Turn signal lamps shall conform to SAE J588. Operating units shall conform to SAE J589 class A and shall be mounted on the steering column. Signal units shall be visible and audible in combination with the towed vehicle and shall be visible when not in combination with the towed vehicle. Turn signal operating units shall have a visible and audible type flash indicator.

3.4.2.6 Semitrailer lighting cable. A semitrailer lighting cable conforming to SAE J1067 and incorporating a connector conforming to SAE J560 on the semitrailer end shall be provided. The cable shall be coiled and shall have an extended length of not less than 110 inches. The SAE J560 connector shall include a grip for withdrawing from the semitrailer receptacle. Stowage for the cable shall be provided by means of hook and hanging loop or protective holding bracket. When hook and loop method is used, the cable shall be furnished with a hanging loop, so attached as to insure that the plug is pointed down when the cable is stowed in the hook provided on the rear of the truck tractor cab or on a pogo stick type hose tender. The protective bracket, when furnished, shall be mounted at the rear of the cab, below the roof line, and shall hold the cable plug within the bracket housing to preclude entrance of weather within the terminals.

3.4.2.7 Trailer lighting cable. A trailer lighting cable, conforming to SAE J1067, shall be furnished for use with the lighting receptacle specified in 3.1.1.7. The cable shall be not less than 110 inches long and both ends of the cable shall be equipped with a round plug conforming to SAE J560. The plugs shall be equipped with a grip for withdrawing from the connector sockets. The cable shall be packaged and stowed in the vehicle tool compartment.

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* 3.4.2.8 Batteries. Each battery shall be of 12-volt potential. The total reserve capacity rating at 80°F and total cold cranking ampere rating at 0°F, both measured in accordance with SAE J537, shall be not less than specified in table II. Batteries shall be of the maintenance free type having the maintenance-free characteristics listed in W-B-131.

TABLE II. Batteries.

Engine	Reserve capacity (minutes)	Cold cranking (amperes)
Diesel	320	1200
* Gasoline	110	450

3.4.2.9 Auxiliary 24-volt system with trailer receptacle. When specified (see 6.2), an auxiliary 24-volt system with a trailer receptacle assembly shall be furnished. Either a converter type (see 3.4.2.9.1) or an alternator type (see 3.4.2.9.2) system meeting specified requirements shall be furnished. A trailer receptacle conforming to MS75021-2, with cover assembly, shall be provided in an accessible location on the rear of the cab. A twelve-conductor truck tractor jumper cable not less than 10 feet long, with both ends of the cable equipped with connectors conforming to MS75020-1 and MS75020-2 shall be furnished. The cable assembly shall be stowed in the vehicle. The 24-volt, 12-ampere output service lighting circuit shall be connected through the appropriate lighting controls to terminals B, D, E, J and L of MS75021-2. A cab mounted hose tender shall be provided to accommodate and secure the 24-volt jumper cable.

3.4.2.9.1 Converter-type 24-volt system. The 12- to 24-volt converter(s) shall operate from the 12-volt battery (see 3.4.2.8). The output capacity shall be not less than 12 amperes.

3.4.2.9.2 Alternator type 24-volt system. The alternator type 24-volt system shall be separate from the 12-volt vehicle lighting and ignition system and shall include:

(a) A nominal 24-volt alternator with not less than 25-ampere rated capacity and capable of providing not less than 7 amperes dc output at normal engine idle speed;
(b) Two 12-volt batteries with a combined capacity of at least 40 ampere-hours at a 20-hour rate or one 24-volt battery with at least 20 ampere-hours capacity at a 20-hour rate;
(c) A voltage regulating device;
(d) An ammeter, mounted on the instrument panel.

3.4.2.10 Radio interference suppression. The vehicle shall be suppressed to limit the electromagnetic radiation in accordance with SAE J551. Any body equipment emitting radiation shall be suppressed to the same level as the vehicle chassis.

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3.4.3 Fuel system. The fuel system shall conform to Federal Motor Carrier Safety Regulations 393.65 and 393.67.

3.4.3.1 Air cleaner. An air cleaner shall be furnished.

* 3.4.3.2 Fuel tank(s). Fuel tank(s) shall be not less than 50-gallon total capacity (43 gallons for vehicles destined for California). When more than one tank is furnished on diesel engine driven vehicles, means shall be provided to assure an equalized fuel level in both tanks. When more than one tank is furnished on gasoline engine driven vehicles, a selector valve, connecting either tank to the engine fuel intake, shall be provided and means shall be provided to monitor the fuel level of either tank from a single fuel gage; or an equalizing pump shall be used to maintain the same level in both tanks.

3.4.4 Exhaust system. The exhaust system shall conform to Federal Motor Carrier Safety Regulation 393.83. The exhaust system, when stacked, shall be furnished with a heat shield and a hinged rain cap. On diesel engine driven vehicles the tail pipe shall be stacked on the curbside of the engine transversely in line with the rear half of the cab. All portions of the exhaust system above the body sheet metal shall be fully enclosed by a heat shield. On gasoline engine driven vehicles, all portions of the exhaust system which project to the rear of the cab shall be provided with a cover plate. The cover plate shall form a shield that will shed fluids, precluding spilled fuel from coming into contact with the exhaust system. A spark arrester shall be furnished, except on turbocharged engines. The spark arrester shall have an 80 percent arresting efficiency when rated in accordance with SAE J350.

* 3.4.5 Automatic transmission. An automatic transmission shall be provided. Input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine. The transmission shall include a hydraulic torque converter and not less than five forward gear ratios. Normal driving range selector position shall provide not less than four forward gear ratios without movement of the selector. The transmission shall be provided with a power takeoff opening. When specified (see 6.2), a hydraulic retarder shall be provided.

3.4.5.1 Power takeoff. A power takeoff shall be provided. The power takeoff shall be of a rated capacity to operate powered equipment. Controls to operate the power takeoff shall be located in the truck cab accessible to the seated driver. A caution plate or decal reading "DO NOT OPERATE VEHICLE AT HIGHWAY SPEEDS WITH POWER TAKEOFF ENGAGED" shall be provided and installed so as to be readily visible to the seated driver.

3.4.6 Driveline components. Driveline components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear reduction. The drive shaft shall be protected in a manner similar to that specified for buses by Federal Motor Carrier Safety Regulation 393.89.

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3.4.7 Frame. Chassis frame shall be the manufacturer's standard for the vehicle furnished. Frame shall have sufficient structural strength and resisting bending moment to at least equal the loads imposed under the specified operating conditions with vehicle loaded to specified GVW. When frame reinforcements are required to meet the specified loads, the reinforcements shall extend at least from the rear of the front-suspension, rear hanger bracket to the front of the rear-spring, front hanger bracket. Crossmembers shall be provided at least rearward of the transmission and at the rear. The chassis frame rails shall be cut off immediately to the rear of the rear spring rear hanger bracket or the frame crossbar closest to the rear of this bracket.

3.4.8 Suspension. The vehicle shall be equipped with a front spring suspension system with components having a rated capacity at least equal to the loads imposed in each member, measured at the ground, with the vehicle loaded to specified GVW. When a spring suspension system is provided for the rear axle, the rear spring suspension system with components shall have a rated capacity at least equal to the load imposed on each member, measured at the ground with the vehicle loaded to the specified GVW. When the suspension system is rated at the spring pads, unsprung weight shall be deducted. Vehicle shall be equipped with hydraulic double-acting shock absorbers at the front wheels.

* 3.4.9 Axles. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, with the vehicle loaded to specified GVW. When specified (see 6.2), the wheel bearings and axle spindles shall be oil lubricated. The hubcaps, except for driving axles, shall have a window for visual determination of oil level. Provisions for venting or withstanding internal pressure buildup and for replenishing the oil supply shall be provided.

3.4.9.1 Traction control. Traction control shall be furnished on the rear axle. The traction control shall actuate automatically to insure that power is transmitted to the wheels having traction when the opposite wheel loses traction. Maximum traction capabilities shall be maintained at all times under each drive wheel for the life of the vehicle.

* 3.4.10 Wheels, rims, tires and tubes. The vehicle shall be equipped with single front and dual rear wheels. Rims and tire ratings shall conform to Tire and Rim Association or European Tyre and Rim Technical Organisation recommendations for the type and size of tires furnished. Multi-piece rims shall not be furnished on the front axles rated at 14,000 pounds or less, and on rear axles rated at 23,000 pounds or less. Tire, rim and disc wheel sizes shall be the same for all wheels on each vehicle. When specified (see 6.2), 10-stud disc type wheels shall be furnished.

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* 3.4.10.1 Tires. Unless otherwise specified (see 6.2), tires shall have highway tread. Tires shall be of the tubeless type or, for axles where multipiece rims are permitted (see 3.4.10), may be of the tube type. Unless otherwise specified, low profile or standard profile tires may be furnished. When specified (see 6.2), low profile tires shall be furnished. Tires shall be of the steel belted radial type or when specified (see 6.2), bias ply. When tube type tires are permitted herein or specified, multipiece rims may be furnished. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with the vehicle loaded to specified GVW. Tires and tires size designation systems shall conform to the Tire and Rim Association or to the European Tyre and Rim Technical Organization recommendations.

* 3.4.10.2 Inner tubes. When tube type tires are permitted herein or specified in procurement documents, inner tubes shall be of heavy duty type and shall be of proper size for tires furnished. Tire flaps shall be provided for tube type tires in accordance with Tire and Rim Association recommendations.

* 3.4.11 Brakes. Brakes shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.43 and 393.45 through 393.52. Brake linings shall be of nonasbestos material on axles rated at 23,000 pounds or less.

3.4.11.1 Service brakes. The vehicle shall be equipped with full-air brakes on all wheels. The braking system, complete with all necessary components, shall include:

(a) Air compressor, unloader-head type, engine driven and engine lubricated, air or water cooled, and having a capacity of not less than 12 cubic feet per minute (cfm);
(b) Air storage reservoir(s), each tank equipped with drain, and with safety and check valves between compressor and last reservoir tank;
(c) Foot control, suspended or treadle type;
(d) Air control valves;
(e) Air pressure gage, visible to the driver;
(f) Low-air-pressure warning, visible and audible;
(g) Service brake stop lamp switch;
(h) Automatic moisture ejector;
* (i) For vehicles with rear axles rated at 23,000 pounds or less, automatic slack adjusters on cam type brakes or internal self-adjusting brakes on wedge and disc type brakes on all axles. Vehicles without this feature on one axle (rear axle rated at over 23,000 pounds) shall not have it on any axle.

3.4.11.1.1 Air dryer. A replaceable cartridge desiccant type air dryer shall be installed in the air brake system. The dryer shall have a capability of removing not less than 95 percent of the moisture in the air being dried. The dryer shall have a pre-cooler and a filter to screen out

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oil and solid contaminants. The dryer shall have an automatic self-cleaning cycle and a thermostatically controlled heater to prevent icing of the purge valve.

3.4.11.2 Trailer brake control system. In addition to the components specified in 3.4.11.1 and 3.4.11.1.1, a trailer brake control system shall be furnished. The trailer brake control system shall include:

(a) Identification of emergency and service lines;
(b) Coincident control of trailer brakes with prime mover foot control;
(c) Independent hand control for trailer brakes;
(d) Prime mover protection valve with dash control and automatic break-away feature;
(e) Trailer stoplight control operative with foot brake and with hand control for trailer brakes;
(f) Two SAE J844 coiled air brake hoses, not less than 110 inches in length when fully extended, equipped with coiled spring hose guards, and SAE J318 gladhand quick connectors on the trailer end of the hoses for towing semitrailers;
(g) Dummy gladhand connectors to retain these hoses when not in use;
(h) Two SAE J844 coiled air hoses, not less than 110 inches long when fully extended, with SAE J318 gladhand couplers on both ends of hoses for towing full trailers. The hoses shall be packaged and stowed in the vehicle tool compartment for shipment;
(i) Air connectors for full trailers with SAE J318 gladhand couplers at the rear of the truck tractor located to preclude interference with the full trailer. Air connectors shall be provided with a cover assembly and a security chain for each connector;
(j) Prime mover only parking brake valve to permit prime mover parking brakes to be applied while charging the trailer air brake system.

3.4.11.3 Brake controls for use from a towing vehicle. When specified (see 6.2), the vehicle shall be furnished with a system for controlling the brakes from a towing vehicle (wrecker). The installation shall be complete with air brake couplers, relay emergency valve with no-bleed-back feature except when spring applied emergency brake is furnished, additional air lines and fittings. The service and emergency couplers shall be mounted on the front in a protected position providing for ready attachment of air hoses from a towing vehicle. The service and emergency couplers shall be identified and provided with dummy gladhand couplers with chains. The system shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard.

3.4.12 Cab. The manufacturer shall furnish a half width cab suited to spotter truck operations. Tilt type half cab, if furnished, shall include a tilting and locking mechanism. Tilt cab features shall be in accordance with the Employee Safety and Health Standards of Federal Motor Carrier Safety Regulation 399, including the first step height, which shall be not more than

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24 inches. The half cab, conventional on tilt type, shall have a sedan type driver's side door and a sedan sliding or folding-leaf type rear door. Doors shall be equipped with locks operable from inside the cab. Driver's side door shall be equipped with an external key operated lock. Drip rails shall be installed above the cab doors, and above windows that open. The window in the door on the driver's side shall be mechanically operated. The curb side window shall be either mechanically operated or shall be of the sliding type. The window in the rear door shall be capable of being opened. Means of opening will be dictated by the type of door supplied. Grab rails and treaded steps shall be provided on both sides of the vehicle to assist personnel in climbing into the cab and onto the truck-tractor deck plate. A suitable step and grab rail shall be provided at the front of the vehicle to facilitate cleaning of the windshield.

3.4.12.1 Cab interior. The cab shall have an upholstered adjustable seat and back for the driver. The seat shall be capable of swiveling 90 degrees to the right, from normal driving position, to allow the driver easy entry and exit by the rear door. The color of the upholstery and the interior finish shall be compatible with the exterior color. White upholstery shall not be furnished. An interior dome light shall be provided. Manufacturer's standard fastenings and one pair of seat belts shall be installed.

3.4.13 Steering. Power steering shall be furnished.

3.4.14 Windshield wipers and washers. The vehicle shall be equipped with a windshield wiper and a washer. The windshield wiper shall be of the multispeed type and operated by either air or an electric motor.

3.4.15 Bumpers. A channel type heavy duty front bumper shall be provided.

3.4.16 Stowage. Stowage space of sufficient size to accommodate a vehicle jack, hand tools, anti-skid chains and emergency reflective triangles shall be furnished for retaining equipment during vehicle operation. Stowage space for these tools may be furnished inside the cab. When stowage space for these tools is located outside the cab, it shall be weatherproof and shall provide for locking with a padlock.

3.4.16.1 Tools. When specified (see 6.2), the vehicle shall be furnished with tools required for exchanging a mounted tire assembly with the spare assembly, and shall include at least a hydraulic jack, jack handle and wheel-nut wrench. The jack shall be of such closed height as to permit its location under axle, or other satisfactory lift point, at any wheel with the tire flat. The jack, without blocking, shall be capable of raising any wheel of the loaded vehicle to a height adequate to permit removal and replacement of the wheel and tire assembly.

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3.4.17 Heater and defroster. The vehicle shall be provided with a hot water heater with fresh air intakes and discharge outlets to floor and to windshield defroster louvers. The heater shall be complete with blower and mounted controls convenient to the driver.

3.4.18 Controls and operating mechanisms. All controls and operating mechanisms shall be located for lefthand drive. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of operator, to and from the driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. All instruments shall be visible to the driver when seated in driving position.

* 3.4.19 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

(a) Key operated ignition switch
(b) Ammeter, charging indicator or voltmeter
(c) Fuel gage
* (d) Oil pressure gage or red indicator warning light
* (e) Engine coolant temperature gage or high coolant temperature or low coolant level red indicator warning light
* (f) High coolant temperature or low coolant level alarm buzzer
(g) Speedometer with recording odometer
(h) Ash receptacle
* (i) Sunvisor
(j) Hydraulic or pneumatic fifth wheel elevating and descending control
* (k) Fifth wheel unlock mechanism located inside cab and accessible to seated driver
(l) Front door mounted armrest on driver's side
(m) Driver's compartment ventilator other than window
(n) Tachometer
* (o) A sensor and load gage, measuring and indicating the load on the fifth wheel.

When specified (see 6.2), an ammeter or voltmeter gage, an oil pressure gage, and a coolant temperature gage shall be provided in lieu of the indicators and buzzers specified above. When specified (see 6.2), an automatic engine shutdown system also shall be furnished. The system shall include an engine coolant temperature, engine coolant level and engine oil pressure red indicator warning light and alarm buzzer. The warning light and buzzer actuation shall precede engine shutdown. The system shall permit engine restart and run for approximately 30 seconds following automatic shutdown. Also, engine oil pressure and engine coolant temperature gages shall be provided with the automatic engine shutdown system.

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* 3.4.20 Rearview mirrors. The vehicle shall be equipped with outside rearview mirrors. The mirrors shall be of the combination type having flat and convex areas. The flat portion shall have not less than 50 square inches of reflective area. The convex area shall have not less than 25 square inches of reflective area and a radius of curvature of not less than 20 inches. One mirror shall be mounted on the street side of the cab. The other shall be pole-mounted in a suitable position on the curbside of the vehicle to afford the driver proper rear vision when moving standard width trailers and semitrailers. Mirrors shall be friction and shockproof mounted and shall be fully adjustable. The streetside mirror shall have not less than two supporting arms.

3.4.21 Horn. Manufacturer's standard electric horn shall be furnished. When specified (see 6.2), in addition, an air operated horn shall be provided.

3.4.22 Engine hour meter. An engine hour meter having a totalizing mechanism of not less than 9,999 hours shall be furnished for the chassis engine to register accurately the number of hours of operating time. The meter shall be of rugged construction to insure continuous trouble free performance under severe operating conditions. The engine hour meter shall be mounted on the instrument panel or in the engine compartment in a readable location.

* 3.4.23 AM/FM radio. When specified (see 6.2), the manufacturer's standard AM/FM radio shall be provided.

* 3.5 Hydraulic lift fifth wheel. Unless otherwise specified (see 3.6), vehicle shall be equipped with a hydraulic lift fifth wheel. The hydraulic fifth wheel unit shall be a power operated, all steel, heavy duty item with a rated capacity to lift and support a load of not less than 50,000 pounds for class 32 and 70,000 pounds for class 44 over the kingpin, and with a lifting range of not less than 14 inches from the lowest to the highest position. The fifth wheel and its mounting structure shall have a design capacity to withstand a drawbar pull of not less than 60,000 pounds for class 32 and 100,000 pounds for class 44 without residual deformation to the mounting, the fifth wheel or the truck tractor structure. The fifth wheel shall be of the fore and aft rocking type with a 36-inch diameter steel coupler plate supplied with beveled approach forks. The fifth wheel unit shall conform to Federal Motor Carrier Safety Regulation 393.70(b).

3.5.1 Fifth wheel subframe. The fifth wheel assembly shall be mounted to a subframe. The subframe (or tractor main frame) shall be provided with an inclined ramp presenting a continuous slope for semitrailer approach. The subframe shall be mounted to the vehicle chassis frame rails. Means shall be provided to prevent deformation of the subframe and the chassis frame rails. The subframe, fifth wheel assembly and its mounting shall be capable of accepting the loads imposed during coupling, uncoupling, ascending and descending ramps up to 14 degrees from horizontal and the effects of weight transfer up and down the specified grades.

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3.5.2 Fifth wheel hydraulic system. The fifth wheel unit shall include a hydraulic system and mechanism arranged to actuate an elevating platform equipped with a fifth wheel; shall incorporate a pneumatic system to open a semiautomatic lock for an SAE J700 kingpin; shall embrace a remote control to permit all fifth wheel operations from within the vehicle cab; and shall include all components, such as pump, hose and fittings, to comprise a complete installation. The remote control shall be provided with a guard to prevent accidental actuation.

3.5.3 Clearance. The clearance from the centerline of the kingpin to the cab, or pogo stick type hose tender when furnished, with the fifth wheel at all elevated positions, shall be not less than 60 inches and not more than 72 inches. The landing wheel clearance from the centerline of the kingpin to a vertical plane on the outside edge of the rear tires, chassis frame rails and rear fenders, with the fifth wheel in the lowest position, shall be not more than 60 inches. Under no condition shall the fifth wheel kingpin locks be positioned to the rear of the centerline of the rear axle. The unladen level height from the ground to the top of the coupler plate of the installed fifth wheel unit with the unit in its lowest position shall be 48 inches, plus or minus 1 inch. Actuation through the full lifting range shall not cause the center of the kingpin lock to shift more than 6.250 inches, measured along the longitudinal centerline of the vehicle.

* 3.5.4 Frame stops. Unless otherwise specified, frame stops shall be provided. Frame stops shall be capable of holding the fifth wheel in an elevated position of 55 inches plus or minus 1 inch, without carrying the imposed load on the lifting cylinders. Stops shall be hinged and shall fold out. A caution plate shall be furnished and mounted in a conspicuous location in the driver's compartment. At least the following information shall be shown: WHEN TOWING SEMITRAILER IN AN ELEVATED POSITION FOR MORE THAN 1,000 FEET, FRAME STOPS ARE TO BE USED TO KEEP IMPOSED LOAD OFF THE LIFTING CYLINDERS. The caution plate shall conform to composition A (class 1 or 2) or composition C of MIL-P-514. When specified (see 6.2), or at the manufacturer's option, the hydraulic fifth wheel lifting cylinder(s) shall be fitted with hydraulically locked, flow-check safety valves in lieu of frame stops. The safety valves shall prevent the fifth wheel being lowered except by hydraulic power and shall prevent shock loading of the hydraulic system.

* 3.5.5 Deck plate. Except for vehicles furnished with saddle tanks, self-cleaning grating of sufficient structural strength for use by the operator in connecting air and electric lines between the tractor and a semitrailer shall be installed. The grating shall extend across and shall be bolted to the frame rails or structural members with sufficient overhang to cover and protect side mounted tanks and fittings. The grating shall be located as close to the cab as possible and shall extend rearward to the maximum extent practicable. Provisions to allow for access by personnel climbing onto the deck plate shall be furnished. Access through the grating for maintenance of fittings and other equipment shall be furnished. The deck plate shall be free of ragged or sharp exposed edges.

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3.5.6 Hose tender. When a tilt type cab is furnished, a pogo stick type hose tender shall be provided behind the cab to accommodate and secure the semitrailer lighting cable and air hoses. When a conventional cab is furnished, a hose tender as specified in 3.4.2.5 may be provided in lieu of the pogo stick. The pogo stick type hose tender shall be mounted on the truck tractor deck plate or chassis members.

* 3.6 Air operated lift fifth wheel. When specified for class 32 vehicles (see 6.2), the vehicle shall be equipped with an air operated lift fifth wheel in lieu of a hydraulic lift fifth wheel. The air operated fifth wheel unit shall be a self-contained, all steel, heavy duty item, with a capacity to lift and support a load of not less than 40,000 pounds over the kingpin, and with a lifting range of not less than 13 inches, from the lowest to the highest position. Actuation through the full lifting range shall not cause the center of the kingpin lock to shift more than 4 inches measured along the longitudinal centerline of the vehicle. The fifth wheel unit shall utilize a base furnished with an incline ramp presenting a continuous slope for a semitrailer approach, and provided with a means for "bolt-on" attachment onto the vehicle chassis frame. The fifth wheel and its mounting structure shall have a design capacity to withstand a drawbar pull of 60,000 pounds without residual deformation to the mounting, fifth wheel or the truck tractor structure. The fifth wheel installation shall conform to 3.5.1.

* 3.6.1 Air operated air lift fifth wheel components. The air operated air lift fifth wheel shall be furnished and installed with at least the following:

(a) Air reservoir tank(s), not less than 7.4 cubic feet total capacity, equipped with drain, safety and check valves between compressor and tank
(b) Air compressor, not less than 12 cfm, air or water cooled, engine driven and engine lubricated, unloader-head type
(c) Automatic moisture ejection valve
(d) Two air starter valves to emit and expel air from the reservoir tank to the air bellows
(e) Three-way valve, for raising, lowering or holding the fifth wheel in any desired position. The valve control shall be mounted on the truck tractor instrument panel in an accessible location for the driver, when in a seated position. The control shall be provided with a guard to prevent accidental activation
(f) Snubbers to eliminate spring deflection
(g) Pneumatic system to open and lock the SAE J700 kingpin, operable from the truck tractor instrument panel
* (h) Manual secondary lock to prevent accidental uncoupling.

3.6.2 Fifth wheel (air operated). The fifth wheel shall be of the fore and aft rocking type, with a 36-inch cast steel coupler plate with beveled approach forks. The fifth wheel assembly shall conform to Federal Motor Carrier Safety Regulations 393.70(b).

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* 3.6.3 Clearances. The fifth wheel clearance measured from the centerline of the kingpin locks to the cab, or pogo stick, with the fifth wheel at all elevated positions, shall be not less than 60 nor more than 68 inches. The landing wheel clearance to the vertical plane of the outside edge of the rear-most tire and the chassis frame rails, with the fifth wheel in its lowest position shall be not more than 64 inches. The unladen level height from ground level to the top of the fifth wheel plate shall be 52 inches plus or minus 1 inch.

3.6.4 Deck plate. A deck plate shall be provided in accordance with 3.5.5.

3.6.5 Hose tender. A hose tender shall be furnished conforming to 3.5.6.

3.7 Servicing and adjusting. Prior to acceptance of the vehicle by the Government inspector, the contractor shall service and adjust each vehicle for operational use including at least the following: alinement of lights; adjustment of the engine, electrical and brake systems; filling and charging of battery; alinement of front wheels; inflation of all tires; cycling the fifth wheel assembly through its entire range of elevations; complete lubrication of chassis, engine, and running gear with grades of lubricants recommended for the ambient air temperature at the delivery point; servicing of cooling system with a solution of ethylene glycol type antifreeze and water in equal parts by volume; and servicing of windshield washer reservoir with water and appropriate additives.

3.8 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Welded, bolted, and riveted construction utilized shall be in accordance with the highest standards of the industry.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examination and tests) as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the

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specification shall not relieve the contractor of the responsibility for ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

4.3 First production vehicle inspection. The first vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformance to the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

4.3.1 Vehicle weight. The first production vehicle shall be weighed to determine curb weight and the distribution of curb weight on the front and rear axles. The imposed loading on front and rear axles shall be computed by the contractor and verified by the Government using the curb weight, the operator weight at 175 pounds, and the payload to provide the specified GVW. Calculated imposed loads on front and rear axles shall be utilized to evaluate the capability of the suspension, axles and tires furnished to meet performance requirements.

4.3.2 Road test. The first production vehicle shall be examined and road tested (less payload) by the contractor. The road test shall be for not less than 10 miles at speeds up to 45 mph.

* 4.3.2.1 Fifth wheel lift test. A semitrailer shall be loaded so that the weight imposed on the fifth wheel when coupled to the vehicle will yield not less than 95 percent of the truck tractor rear axle rating. The spotter shall perform 50 continuous cycles of coupling to the semitrailer, lifting through the full range of fifth wheel movement, then lowering and uncoupling. Following the test, the vehicle and all components shall be inspected for damage, deformation and malfunction.

* 4.3.2.2 Alternate fuel test. When operational capability on JP-4, JP-5 and JP-8 is specified, the first production vehicle shall be tested. The contractor shall obtain a sufficient amount of JP-4, JP-5 and JP-8 fuel for

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the following tests. When changing fuel types, the fuel tank and filters shall be drained, the new fuel substituted, and sufficient idling or driving shall be performed to purge the previous fuel type from the system. Using JP-4, the engine shall be warmed up for not less than 15 minutes. Typical yard spotting maneuvers shall be performed for a period of not less than 30 minutes using a semitrailer loaded to provide full GCW. The test shall be repeated with JP-5. Using JP-8 fuel, the engine shall be warmed up for not less than 15 minutes. The yard spotting maneuvers shall again be performed, but for a one hour period, followed by a road test of not less than 60 miles at speeds up to maximum geared speed, at full GVW.

4.3.3 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample, and shall be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

4.3.4 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under contract until corrective action has been taken.

4.4 Inspection of production vehicles. The contractor's inspection system shall, as a minimum, assure that the vehicle conforms to the physical and dimensional requirements and is capable of meeting performance requirements contained herein. For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance, records acceptable to the Government indicating that the servicing and adjusting required by 3.7 have been accomplished.

5. PREPARATION FOR DELIVERY

5.1 Vehicle processing. The vehicle shall be processed for shipment, from the manufacturer's plant to initial receiving activity, in accordance with the manufacturer's standard commercial practice.

6. NOTES

6.1 Intended use. The vehicles covered by this specification are intended for general nontactical use by the Government for shuttling and spotting full trailers and semitrailers at depot and roll-on roll-off operations.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
* (b) Class of vehicle required (see 1.2).

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(c) Issues of DODISS to be cited in the solicitation and if required, the specified issues of individual documents referenced (see 2.1.1 and 2.2).
* (d) Identification of appropriate military service for painting (see 3.1.1.1).
* (e) Identification of appropriate military service for marking (see 3.1.1.2).
(f) Concealed marking, if required (see 3.1.1.2).
(g) Rustproofing, if required (see 3.1.1.3).
(h) Tropical rustproofing, if required (see 3.1.1.2).
* (i) Satisfactory operation on JP-4, JP-5 and JP-8 fuels, if required (see 3.4.1.1).
(j) Gasoline engine, if required (see 3.4.1.2).
(k) Silicone rubber hoses, if required (see 3.4.1.7).
(l) Power plant heaters and fuel warmer, if required (see 3.4.1.8).
(m) If a battery heater is not required (see 3.4.1.8).
(n) Auxiliary 24-volt system with trailer receptacle, if required (see 3.4.2.8).
(o) Hydraulic retarder, if required (see 3.4.5).
* (p) Oil lubricated wheel bearings and axle spindles, if required (see 3.4.9).
(q) Disc type wheels, if required (see 3.4.10).
* (r) Bias ply, low profile or other type tires, if required (see 3.4.10.1).
(s) Brake controls for use from a towing vehicle, if required (see 3.4.11.3).
(t) Tools, if required (see 3.4.16.1).
(u) Load gage, if required (see 3.4.19(o)).
* (v) Gages if required in lieu of indicators (see 3.4.19).
* (w) Automatic engine shutdown system, if required (see 3.4.19).
(x) Air operated horn, in addition, if required (see 3.4.21).
* (y) AM/FM radio, if required (see 3.4.23).
(z) Hydraulic fifth wheel safety check valve, if required (see 3.5.4).
(aa) Air operated lift fifth wheel for class 32 vehicles, if required (see 3.6).

* 6.3 Performance prediction. SAE Truck Ability Prediction Procedure computations and computations for low speed and maximum geared speed will be required by the contractor. The SAE Work Sheet Item 1 should include vehicle model number and engine model number. Unless other conditions are cited in the contract, computations should be made for normal atmospheric pressure, normal ambient air temperature, and still, dry air. The factors to be sued in predicting truck ability (see 3.3.1.1) are established as follows for the corresponding SAE Truck Ability Prediction Procedure Tables:

Table 1 - Tire Factor. This factor must relate to the size of tires furnished by the contractor in accordance with this specification.

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Table 2	-	Altitude Factor. 1.00.
Table 3	-	Rolling Factor. 1.003 for size 32, 0.278 for size 44.
Table 4	-	Area Factor. 0.225.
Table 5	-	Velocity Factor. 85.7 for size 32, 3.46 for size 44.
Table 6	-	Altitude Factor. 1.00.
Table 7	-	Chassis Friction Horsepower. Use applicable power unit GVW (to nearest, higher, 1,000 pounds) and the engine rpm (to nearest 100 revolutions) which is required for 35 mph geared speed for size 32 and 12 mph for size 44. For GVW and engine speed beyond the range of this table, factors shall be extrapolated.
Table 8	-	Grade Factor. 1.07.
Table 8A	-	Correction Factor. Not required.
Table 9	-	Road Factor. 0.0.

6.4 Subject term (key word) listing.

Air operated fifth wheel
Deck plate
Hose tender
Hydraulic fifth wheel
Shuttle tractor.

6.5 Identification of changes. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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